

# Using Generative AI in Process Workflow

## 1. Introduction

This document includes design, implementation, successes and challenges that are found during the development of automated service request process using Google Gemini within ITIL service request management.

## 2. Project Goal

The goal of project was to automate the IT service request by using generative AI where incoming email trigger the workflow, classify the request and suggest an action and user receive a reply.

## 3. Tools

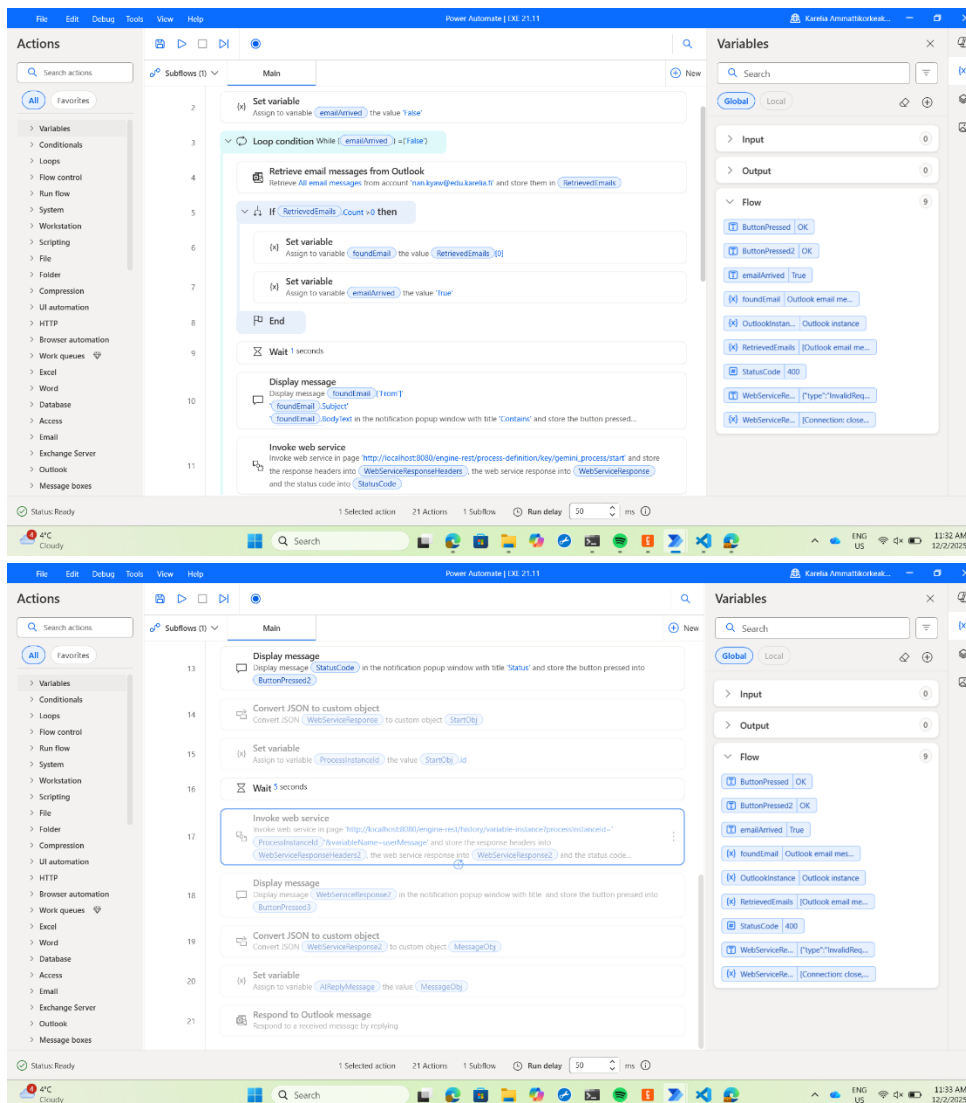
- Power Automate Desktop for receiving and sending emails
- Camunda 7 and Camunda run for workflow
- Node.js for executing external tasks
- Google Gemini API

## 4. Implementation

### 4.1 Power Automate Desktop (PAD)

In this project, PAD perform two main tasks:

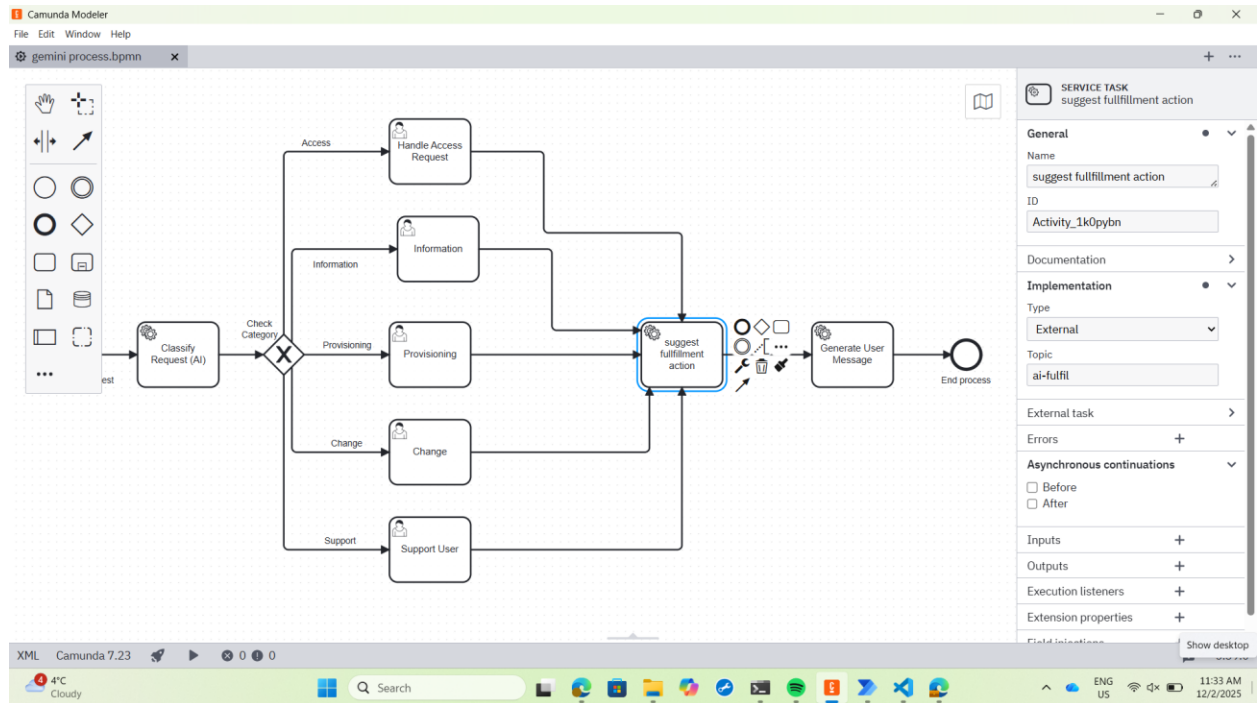
- PAD monitor the outlook inbox and extract sender, subject and message body and trigger the Camunda process by calling REST API
- Receive AI generated and reply to the original sender



## 4.2 Camunda Workflow

The BPMN model include the following steps:

- The request text is classified by using AI and output can be Access, Information, Provisioning, Change or Support
- Based on classification result by AI, the process goes through the branches
- Gemini generate message that will email back to the user

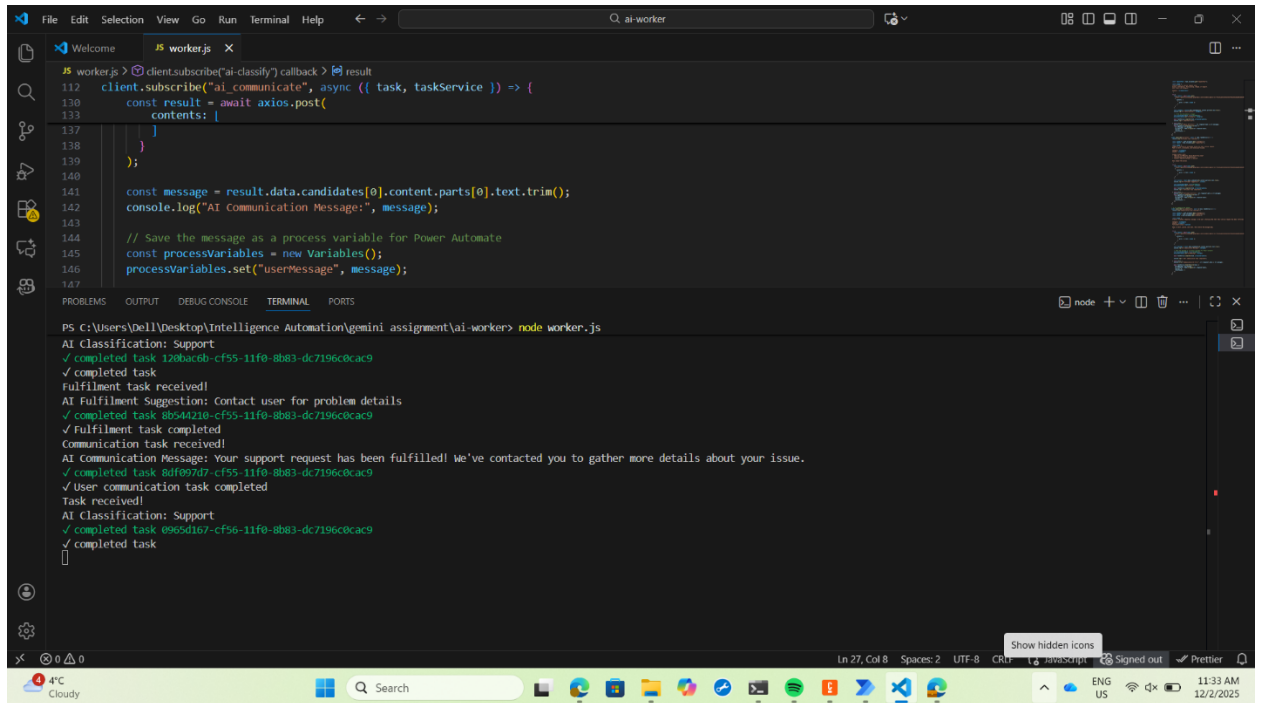


## 4.3 Node.js

Node.js calls Gemini API and handles classification of the service request, generate action for fulfil the request and create response to reply to the sender.

## 5. What Worked Successfully

- PAD successfully monitored outlook inbox, Incoming emails were detected automatically and sender, subject and text body was extracted successfully.
- PAD automatically start Camunda process without manual input
- The AI classification were executed correctly into categories.
- The exclusive gateway in BPMN model routed the process correctly based on classification result.
- The fulfillment suggestion generated by Gemini matched the request category and context.
- AI generated text to reply to original sender after fulfill their request.



```
File Edit Selection View Go Run Terminal Help
ai-worker

Welcome worker.js X
worker.js > client.subscribe("ai-classify") callback > result
112 client.subscribe("ai_communicate", async ({ task, taskService }) => {
130   const result = await axios.post(
133     contents: {
137       }
138     }
139   });
140
141   const message = result.data.candidates[0].content.parts[0].text.trim();
142   console.log("AI Communication Message:", message);
143
144   // Save the message as a process variable for Power Automate
145   const processVariables = new Variables();
146   processVariables.set("userMessage", message);
147 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\ DELL\Desktop\Intelligence Automation\gemini assignment\ai-worker> node worker.js
AI Classification: Support
✓ completed task 120bac60-cf55-11f0-8b83-dc7196c0cac9
✓ completed task
Fulfilment task received!
AI Fulfilment Suggestion: Contact user for problem details
✓ completed task 8b544210-cf55-11f0-8b83-dc7196c0cac9
✓ Fulfilment task completed
Communication task received!
AI Communication Message: Your support request has been fulfilled! We've contacted you to gather more details about your issue.
✓ completed task 8df097d7-cf55-11f0-8b83-dc7196c0cac9
✓ User communication task completed
Task received!
AI Classification: Support
✓ completed task 0995d167-cf56-11f0-8b83-dc7196c0cac9
✓ completed task
[]
```

## 6. Problems Encountered

- Json body that PAD send to Camunda was the string of code instead of the body text of the email.
- PAD reply to original sender before getting message form AI and reply blank email.
- Camunda workflow is not automatically end the process and it was stuck at the generate user message.

## 7. Conclusion

The project demonstrates how generative AI can be integrated into IT service request workflow. Even though several technical challenges are still in process, most of the core functionalities were integrated successfully.